

A Noncooperative Game-Theoretic Framework for Radio Resource Management in 4G Heterogeneous Wireless Access Networks

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Abstract—Fourth generation (4G) wireless networks will provide high-bandwidth connectivity with quality-of-service (QoS) support to mobile users in a seamless manner. In such a scenario, a mobile user will be able to connect to different wireless access networks such as a wireless metropolitan area network (WMAN), a cellular network, and a wireless local area network (WLAN) simultaneously. We present a game-theoretic framework for radio resource management (that is, bandwidth allocation and admission control) in such a heterogeneous wireless access environment. First, a noncooperative game is used to obtain the bandwidth allocations to a service area from the different access networks available in that service area (on a long-term basis). The Nash equilibrium for this game gives the optimal allocation which maximizes the utilities of all the connections in the network (that is, in all of the service areas). Second, based on the obtained bandwidth allocation, to prioritize vertical and horizontal handoff connections over new connections, a bargaining game is formulated to obtain the capacity reservation thresholds so that the connection-level QoS requirements can be satisfied for the different types of connections (on a long-term basis). Third, we formulate a noncooperative game to obtain the amount of bandwidth allocated to an arriving connection (in a service area) by the different access networks (on a short-term basis). Based on the allocated bandwidth and the capacity reservation thresholds, an admission control is used to limit the number of ongoing connections so that the QoS performances are maintained at the target level for the different types of connections.

Index Terms—Heterogeneous wireless networks, bandwidth allocation and admission control, noncooperative game, network utility.

1 INTRODUCTION

ONE of the most important features of the evolving fourth generation (4G) wireless networks will be heterogeneous wireless access [1] in which a mobile will be able to connect to several wireless access networks (for example, wireless local area network (WLAN), cellular network, and wireless metropolitan area network (WMAN)) simultaneously. This would facilitate high-speed wireless Internet access for services such as multimedia streaming from a fixed server using multiple threads at the application layer. In such a heterogeneous wireless access environment, the radio resource management (RRM) mechanisms for bandwidth allocation, congestion control, and admission control must be designed to satisfy both the mobile users' and the service providers' requirements. After all, a mobile user will be able to connect to the Internet in a seamless manner and also the wireless resources need to be managed efficiently from the service providers' point of view.

We present a game-theoretic RRM framework for wireless access in a heterogeneous network environment. The objectives of the proposed framework are to maximize network utility through efficient resource allocation, achieve prioritization among different types of connections such as new connections and vertical and horizontal handoff connections, and ensure that the performance of

ongoing connections does not deteriorate due to accepting too many connections in a service area.

We partition the problem of RRM into two major parts, that is, network-level and connection-level RRM. In the network level, the limited available bandwidth of each network must be allocated to each service area so that the utilities of the different access networks, which are presumably operated by the different service providers, are maximized. In this competitive environment, therefore, we use a noncooperative game to obtain the solution of the bandwidth allocation by the different access networks in a service area. Again, for seamless mobility across the service areas, a portion of the radio resources in a service area needs to be reserved for handoff connections. Since new connections and handoff connections have to share the available bandwidth in a service area, an agreement on bandwidth reservation can be made so that the desired quality-of-service (QoS) performances (for example, handoff connection dropping probability or new connection blocking probability) can be achieved. Therefore, we formulate a bargaining game to obtain the capacity (that is, bandwidth) reservation thresholds for the different types of connections. Both network-level bandwidth allocation and capacity reservation can be performed on a long-term basis based on the average (or steady state) connection-level statistics in the different service areas.

On the other hand, connection-level bandwidth allocation must be performed on a short-term basis and should be adapted upon the arrival and departure of every connection in a service area. Again, each network (that is, service provider) in a service area aims at maximizing its utility while offering bandwidth to a new connection. Therefore, a

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connection-level noncooperative game is solved to obtain the offered bandwidth to the new connection.

Although the problem of bandwidth allocation and admission control in a homogeneous wireless network was extensively studied in the literature (for example, in [2], [3], [4], and [5]), it has not been investigated thoroughly in a heterogeneous wireless access setting considering both the user-centric and network-centric viewpoints. In a heterogeneous environment, different networks could be operated by different service providers. These service providers are rational to maximize their revenues. Therefore, game-theoretic models are useful in the modeling and analysis of resource management protocols in this competitive scenario.

The rest of this paper is organized as follows: Related works are reviewed in Section 2. The model for the heterogeneous wireless access network and the components of the proposed RRM framework are described in Section 3. Section 4 presents the network-level bandwidth allocation and the capacity reservation mechanisms in the proposed RRM framework. The connection-level allocation and admission control mechanisms are described in Section 4. Section 5 presents the performance evaluation results. Conclusions are stated in Section 6.

2 RELATED WORK

The issues related to integration of diverse wireless access technologies such as cellular, WLAN, and mobile ad hoc networks (MANET) and provisioning of QoS to the mobile users were studied in [6]. In [7], an adaptive transport layer (ATL) was proposed for heterogeneous wireless networks with the capabilities of adaptive congestion control, multimedia support, and providing fairness of transmission. In [8], a network selection mechanism for 4G wireless networks was proposed. Specifically, *gray relation analysis* was used to decide which network should be used for each mobile. This decision is based on users' preferences, application requirements, and network conditions. However, the problem of bandwidth allocation was not considered.

Recently, game theory has been widely used for resource management in wireless networks. In [9], the admission and rate control problem for CDMA systems was formulated as a noncooperative game. The formulation considered the choice of a user to churn from a current provider to another. The decision on whether a new user can be admitted or not and the allocated transmission are determined from the Nash equilibrium. An admission control game for CDMA networks was formulated in [10] to obtain an efficient and fair resource allocation for multiple classes of traffic. Game theory was also used to solve the power control problem in wireless networks [11], [12], [13]. However, all these works considered the RRM problem in a single wireless access network.

In [15], a cooperative game framework was proposed for bandwidth allocation in 4G heterogeneous wireless access networks. The bandwidth allocation problem was formulated as a cooperative game (that is, bankruptcy game) and the solution (that is, the amount of bandwidth offered to a new connection) was obtained from Shapley value. This is different from a noncooperative approach in which each network is rational and selfish to maximize its own profit. In a noncooperative environment, all networks compete

with each other to achieve their objectives. More specifically, the difference between cooperative and noncooperative approaches lies in the fact that the former is group oriented, whereas the latter is individual oriented. In a cooperative approach, groups of players seek fair resource allocation. On the other hand, in a noncooperative approach, allocation is performed based on the individual's profit gained from the resource.

In a heterogeneous wireless access environment, a vertical handoff mechanism needs to consider not only the radio link and the physical layer parameters but also the network and the transport layer parameters. In [16], a framework for vertical handoff was presented where the handoff decision metrics include service type, data rate requirement, network condition, and cost of handoff. A dynamic optimization was proposed to provide a QoS guarantee to the mobile users while maximizing the network utilization. In [17], a mobility management solution was proposed for heterogeneous wireless access networks to handle vertical handoff and network roaming. This solution was designed based on a formal policy representation model for a decision-making process for internetwork mobility. However, maximization of network utility (from service providers' point of view) was not considered in these works.

The problem of integrating WLANs into the cellular wireless networks was investigated in the literature. In [18], a hierarchical RRM framework was designed to support seamless handoff between a WLAN and a cellular network. A hierarchical and distributed framework for seamless roaming across cellular networks and WLANs was proposed in [19]. The QoS mapping and Internet work message translation mechanisms were designed to support seamless handoff among multiple WLANs and cellular networks. However, a more general heterogeneous network architecture should be considered for RRM in 4G wireless networks.

3 MODEL FOR HETEROGENEOUS WIRELESS ACCESS AND THE RRM FRAMEWORK

3.1 System Model

We consider a heterogeneous wireless access environment consisting of IEEE 802.11 WLAN, CDMA cellular network, and IEEE 802.16 WMAN radio interfaces, as shown in Fig. 1. A mobile with multiple radio transceivers (for example, software radio) is able to connect to these radio access networks simultaneously.

We consider a geographic region that is entirely covered by a WMAN base station and partly covered by cellular base stations and partly by WLAN access points (APs), as shown in Fig. 1. Users in the different service areas in this region have access to different types and different numbers of wireless networks. In particular, in area 1, only WMAN service is available. In area 2 and area 4, services from cellular networks and WMAN are available. In area 3 and area 5, a mobile can connect to all three types of networks. Note that the RRM framework to be presented later in this paper can be applied to any other service area setting (different from the one shown in Fig. 1) in a considered geographic region.

Different wireless access networks are operated by different service providers. We assume that a mobile is

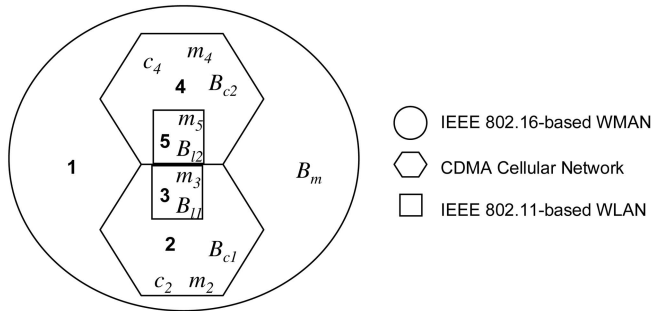


Fig. 1. Service areas under consideration in a heterogeneous wireless access environment.

able to connect to each of the networks in the corresponding service area and perfect power control is assumed to ensure a uniform available transmission rate across the coverage area. A connection can be handed over between service areas with same types of wireless networks (for example, from area 2 to area 4, both of which have services available from WMAN and cellular network) or between areas with different types of networks (for example, from area 1 to area 2). We refer to the former as a horizontal handoff, whereas the latter is a vertical handoff.

In this heterogeneous wireless access network, we assume that multi-interface mobile terminals are able to connect to three different wireless access networks simultaneously. These wireless access networks are IEEE 802.11 WLAN, CDMA cellular network, and IEEE 802.16 WMAN.

3.1.1 IEEE 802.16-Based WMAN

We consider an IEEE 802.16-based WMAN radio interface operating at 10-66 GHz band that supports data rate in the range of 10-50 Mbps depending on the bandwidth of operation and the modulation and the coding schemes [22]. The transmission frame is divided into subframes for downlink and uplink transmissions. Although time-division multiplexing (TDM) is used for downlink transmission, time-division multiple access (TDMA) is used for uplink transmission. These subframes are composed of transmission bursts (that is, uplink and downlink bursts), which carry MAC information and users' data. Each transmission burst corresponding to a particular mobile, which contains several MAC layer packet data units (PDUs), is separated from the other by a preamble field.

3.1.2 CDMA-Based Cellular Wireless Access

We consider a wideband CDMA cellular wireless access system [21]. For a certain number of active users in a cell, the signal-to-interference-plus-noise ratio (SINR) for a user for uplink transmission is determined from the transmit power of all other mobiles, background noise power at the base station, and intercell interference power. A certain transmission rate for a user can be achieved given the bit-error-rate requirement and its corresponding SINR.

3.1.3 IEEE 802.11-Based WLAN

We consider an IEEE 802.11 WLAN radio interface with distributed coordination function (DCF)-based medium access control (MAC), which uses a carrier sense multiple access with collision avoidance (CSMA/CA) protocol. However, since CSMA/CA is a contention-based MAC

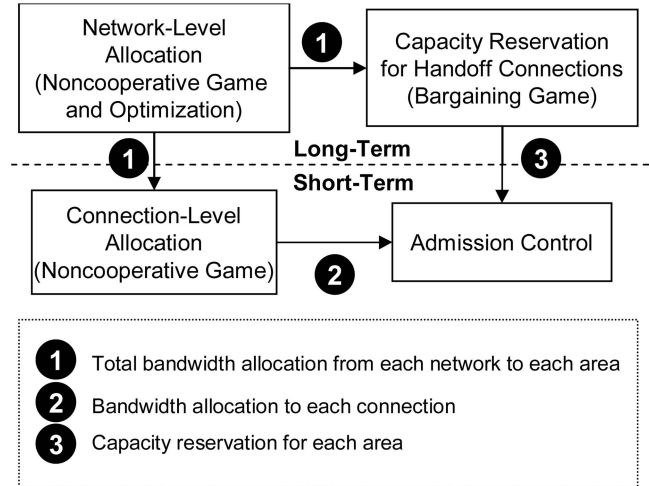


Fig. 2. Components of the proposed RRM framework.

protocol, we adopt a distributed reservation-based MAC protocol, namely, *early backoff announcement (EBA)* [20], which is an enhanced version of DCF. EBA is also backward compatible with IEEE 802.11 DCF. In EBA, by incorporating the backoff information in the MAC header, mobiles can completely avoid collisions.

3.2 RRM Framework

The RRM framework is composed of four components: network-level bandwidth allocation for a service area, capacity reservation for the different types of connections, connection-level bandwidth allocation, and admission control (Fig. 2). In network-level bandwidth allocation, the available bandwidth from different access networks is assigned to the service areas so that all of the service providers are satisfied with the allocation.

Based on the network-level bandwidth allocation, capacity reservation is used for service differentiation among new connections and vertical and horizontal handoff connections. In connection-level allocation, the required amount of bandwidth is allocated to an arriving connection in a service area from the different available access networks. Finally, the admission control component utilizes the results of capacity reservation together with connection-level allocation to decide whether an incoming connection can be accepted or not. Note that, although network-level bandwidth allocation and capacity reservation can be performed in an offline manner (on a long-term basis), connection-level allocation and admission control are performed in an online manner (on a short-term basis) upon the arrival and departure of connections in a service area.

In the system model under consideration, there are multiple service providers who noncooperatively offer wireless access services to the users. In particular, these service providers aim to maximize their utility while allocating bandwidth to the connections. Since each of the service providers has his own interest to maximize his payoff, the bandwidth allocation problem can be modeled as a noncooperative game the solution of which satisfies all of the service providers. Therefore, for network-level bandwidth allocation, assuming that the network service providers are rational, a noncooperative game is formulated and the solution is obtained from the Nash equilibrium

(which will be shown to maximize the total network utility as well).

On the other hand, the problem of capacity reservation among new, horizontal handoff, and vertical handoff connections can be formulated as a resource-sharing problem. In this case, a negotiation among the players can be performed to achieve efficient and fair sharing. This negotiation is practical in this case since all of the connections (new, horizontal, and vertical handoff connections) are in the same service area. Therefore, a bargaining game formulation is used to obtain the fair allocation. In this bargaining game, new connections, vertical handoff, and horizontal handoff connections negotiate with each other to obtain the reservation thresholds so that the connection-level QoS requirements (that is, new connection blocking and handoff connection dropping probabilities) for the different types of connections are satisfied. In this case, we consider the equilibrium as the solution of the game. Both network-level bandwidth allocation and capacity reservation are performed based on the average number of ongoing connections in a service area that can be obtained from the network statistics at the steady state (that is, long-term parameters).

For connection-level allocation, the problem of bandwidth allocation to an incoming connection in a service area is modeled as a trading market. In a service area, each network is assumed to be rational and self-interested to maximize its profit. All networks noncooperatively seek for the optimal strategy so that their profits are maximized. Also, these strategies must be stable in the sense that every firm is satisfied with the solution given other firms' strategies. We establish a revenue function for a network offering bandwidth to an arriving connection by considering the utility that the network gains through allocation of bandwidth to the new connection and the cost function, which accounts for the loss of utility due to revocation of some bandwidth from the ongoing connections.

We formulate a noncooperative game, and the Nash equilibrium is obtained as the solution of this game (that is, the amount of bandwidth offered by each network to an incoming connection). We present two algorithms to obtain the solution, that is, the search and the iterative algorithms. For the search algorithm, an optimization problem is formulated which is solved by a standard direct search method. On the other hand, the iterative algorithm takes advantage of the problem structure to obtain the solution iteratively. The rate of convergence of these algorithms is studied. We conduct an extensive performance evaluation for the proposed RRM framework.

4 NETWORK-LEVEL BANDWIDTH ALLOCATION AND CAPACITY RESERVATION

4.1 Network-Level Bandwidth Allocation

The objective of network-level bandwidth allocation is to allocate bandwidth to a particular service area from each of the available networks in that service area so that all of the service providers are satisfied. This network-level allocation is required to ensure that a particular amount of bandwidth is reserved for each service area that is not affected by sudden traffic fluctuations in other service areas. To analyze network utility in a service area for the above bandwidth

allocation, we use a utility function of throughput defined as follows [14]:

$$U_{i,x}^{(con)} = w \log(\alpha b), \quad (1)$$

where $U_{i,x}^{(con)}$ is the utility of network i for an allocated bandwidth of b to connection x , w , and α are constants indicating the scale and the shape of the utility function.

We formulate a noncooperative game among the different access networks and the Nash equilibrium for pure strategy is obtained as the solution of this game. Then, we show that the stable solution (that is, Nash equilibrium) thus obtained also maximizes the long-term total utility of the entire network. For this, we require the average number of ongoing connections (in the steady state) in the different service areas that can be obtained either analytically or empirically. For simplicity, we use an M/M/m/m queueing model to obtain the average number of ongoing connections in a service area. In particular, given the average connection arrival rate λ (that is, the sum of new connection and handoff connection arrival rates) and the average connection holding time $1/\mu$, the average number of ongoing connections can be obtained from

$$N = \sum_{i=1}^T i \left(\frac{(\rho)^i / i!}{\sum_{j=1}^T (\rho)^j / j!} \right), \quad (2)$$

where $\rho = \frac{\lambda}{\mu}$ and T is a large number (for example, $T = 1,000$).

For the service areas shown in Fig. 1, there are five wireless access networks, that is, one WMAN, two cellular networks, and two WLANs. Let the total available bandwidth in these networks be B_m , B_{c1} , B_{c2} , B_{l1} , and B_{l2} , respectively. Let N_i ($i \in \{1, \dots, 5\}$) denote the average number of ongoing connections in area i , let m_i denote the amount of bandwidth offered by WMAN to area i ($\sum_{i=1}^5 m_i = B_m$), let c_2 and c_3 denote bandwidth offered by cellular network to area 2 and area 3 ($c_2 + c_3 = B_{c1}$), and let c_4 and c_5 denote bandwidth offered by cellular network to area 4 and area 5 ($c_4 + c_5 = B_{c2}$), respectively. Note that B_{l1} and B_{l2} denote the amount of bandwidth available from WLANs in area 3 and area 5, respectively.

4.1.1 Noncooperative Game for Network-Level Bandwidth Allocation

The formulation of the noncooperative game can be described as follows:

- **Players.** The two players of this game are the WMAN and the cellular network for a service area in Fig. 1.¹
- **Strategies.** The strategies for the WMAN are m_i ($i \in \{2, \dots, 5\}$) and those for the cellular network are c_j ($j \in \{2, 4\}$) (Fig. 1).
- **Payoffs.** The payoff for the WMAN is the utility gained from offering bandwidth m_i to all service areas and that for the cellular network is the utility gained from offering bandwidth c_2 , c_3 , c_4 , and c_5 to service areas 2, 3, 4, and 5, respectively.

1. Since a WLAN covers only one service area, WLANs are not considered in this game.

Specifically, the payoff for the WMAN is given as follows:

$$\begin{aligned}
 U_{wman}(m_i, c_j) = w & \left[\left(N_1 \log \left(\alpha \frac{B_m - \sum_{i=2}^5 m_i}{N_1} \right) \right) \right. \\
 & + \left(N_2 \log \left(\alpha \frac{c_2 + m_2}{N_2} \right) - N_2 \log \left(\alpha \frac{c_2}{N_2} \right) \right) \\
 & + \left(N_3 \log \left(\alpha \frac{B_{l1} + B_{c1} - c_2 + m_3}{N_3} \right) \right. \\
 & \left. - N_3 \log \left(\alpha \frac{B_{l1} + B_{c1} - c_2}{N_3} \right) \right) \\
 & + \left(N_4 \log \left(\alpha \frac{c_4 + m_4}{N_4} \right) - N_4 \log \left(\alpha \frac{c_4}{N_4} \right) \right) \\
 & + \left(N_5 \log \left(\alpha \frac{B_{l2} + B_{c2} - c_4 + m_5}{N_5} \right) \right. \\
 & \left. \left. - N_5 \log \left(\alpha \frac{B_{l2} + B_{c2} - c_4}{N_5} \right) \right) \right] \quad (3)
 \end{aligned}$$

and the payoff for the cellular networks is given as follows:

$$\begin{aligned}
 U_{cell}(m_i, c_j) = w & \left[\left(N_2 \log \left(\alpha \frac{c_2 + m_2}{N_2} \right) - N_2 \log \left(\alpha \frac{m_2}{N_2} \right) \right) \right. \\
 & + \left(N_3 \log \left(\alpha \frac{B_{l1} + B_{c1} - c_2 + m_3}{N_3} \right) \right. \\
 & \left. - N_3 \log \left(\alpha \frac{B_{l1} + B_{c1} + m_3}{N_3} \right) \right) \\
 & + \left(N_4 \log \left(\alpha \frac{c_4 + m_4}{N_4} \right) - N_4 \log \left(\alpha \frac{m_4}{N_4} \right) \right) \\
 & + \left(N_5 \log \left(\alpha \frac{B_{l2} + B_{c2} - c_4 + m_5}{N_5} \right) \right. \\
 & \left. \left. - N_5 \log \left(\alpha \frac{B_{l1} + B_{c2} + m_5}{N_5} \right) \right) \right] \quad (4)
 \end{aligned}$$

Here, the payoff for an access network is determined from the surplus utility on that due to the allocations by other networks.

The *Nash equilibrium* of a game is a strategy profile (list of strategies—one for each player) with the property that no player can increase his payoff by choosing a different action given the other players' actions [26]. The pure strategy pair (m_i, c_j) is a *Nash equilibrium* if

$$U_{wman}(m_2^*, m_3^*, m_4^*, m_5^*, c_2^*, c_4^*) \geq U_{wman}(m_2, m_3, m_4, m_5, c_2^*, c_4^*) \quad \forall m_2, m_3, m_4, m_5, \quad (5)$$

$$U_{cell}(m_2^*, m_3^*, m_4^*, m_5^*, c_2^*, c_4^*) \geq U_{cell}(m_2^*, m_3^*, m_4^*, m_5^*, c_2, c_4) \quad \forall c_2, c_4. \quad (6)$$

To determine the *Nash equilibrium*, we use *best response functions* for both players. The best response function for the WMAN at the network level $BR_{wman}^{(net)}(c_2', c_4')$, given that the cellular network chooses strategy (c_2', c_4') , is obtained by finding strategy m_i that maximizes the utility of the WMAN, that is,

$$\begin{aligned}
 (m_2, m_3, m_4, m_5) = \\
 \arg \max_{m_2, m_3, m_4, m_5} U_{wman}(m_2, m_3, m_4, m_5, c_2', c_4'). \quad (7)
 \end{aligned}$$

Similarly, the best response function for the cellular network (at the network level) $BR_{cell}^{(net)}(m_2', m_3', m_4', m_5')$, given that the WMAN chooses strategy (m_2', m_3', m_4', m_5') , is expressed as follows:

$$(c_2, c_4) = \arg \max_{c_2, c_4} U_{cell}(m_2', m_3', m_4', m_5', c_2, c_4). \quad (8)$$

The set of strategies $(m_2^*, m_3^*, m_4^*, m_5^*, c_2^*, c_4^*)$ is a *Nash equilibrium* if and only if $(m_2^*, m_3^*, m_4^*, m_5^*) = BR_{wman}^{(net)}(c_2^*, c_4^*)$ and $(c_2^*, c_4^*) = BR_{cell}^{(net)}(m_2^*, m_3^*, m_4^*, m_5^*)$.

To obtain the best response function for WMAN at the network level, we differentiate U_{wman} with respect to m_2 , m_3 , m_4 , and m_5 assuming that c_2 and c_4 are constants. Then, we have

$$\begin{aligned}
 \frac{\partial U_{wman}(m_i, c_j)}{\partial m_2} = 0 \quad \Rightarrow \\
 N_1(m_2 + c_2) = N_2(B_m - m_2 - m_3 - m_4 - m_5), \quad (9)
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial U_{wman}(m_i, c_j)}{\partial m_3} = 0 \quad \Rightarrow \\
 N_1(m_3 + B_{c1} - c_2 + B_{l1}) = N_3(B_m - m_2 - m_3 - m_4 - m_5), \quad (10)
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial U_{wman}(m_i, c_j)}{\partial m_4} = 0 \quad \Rightarrow \\
 N_1(m_4 + c_4) = N_4(B_m - m_2 - m_3 - m_4 - m_5), \quad (11)
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial U_{wman}(m_i, c_j)}{\partial m_5} = 0 \quad \Rightarrow \\
 N_1(m_5 + B_{c2} - c_4 + B_{l2}) = N_5(B_m - m_2 - m_3 - m_4 - m_5), \quad (12)
 \end{aligned}$$

and for the cellular network, we have

$$\begin{aligned}
 \frac{\partial U_{cell}(m_i, c_j)}{\partial c_2} = 0 \quad \Rightarrow \\
 N_2(B_{l1} + B_{c1} - c_2 + m_3) = N_3(c_2 + m_2), \quad (13)
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial U_{cell}(m_i, c_j)}{\partial c_4} = 0 \quad \Rightarrow \\
 N_4(B_{l2} + B_{c2} - c_4 + m_5) = N_5(c_4 + m_4). \quad (14)
 \end{aligned}$$

By solving the above system of equations (that is, (9)-(14)), we can obtain the solution of the game that is the *Nash equilibrium* and satisfies both the WMAN and the cellular network.

4.1.2 Optimization of Total Network Utility

The total utility of the entire network (that is, the utility for connections in the entire service region) can be obtained from

$$\begin{aligned}
 U_{tot}(m_i, c_i) = w & \left[N_1 \log \left(\alpha \frac{m_1}{N_1} \right) + N_2 \log \left(\alpha \frac{m_2 + c_2}{N_2} \right) \right. \\
 & + N_3 \log \left(\alpha \frac{m_3 + c_3 + B_{l1}}{N_3} \right) \\
 & + N_4 \log \left(\alpha \frac{m_4 + c_4}{N_4} \right) \\
 & \left. + N_5 \log \left(\alpha \frac{m_5 + c_5 + B_{l2}}{N_5} \right) \right] \quad (15)
 \end{aligned}$$

To maximize total utility, an optimization problem is formulated as follows:

$$\text{Maximize : } U_{tot}(m_i, c_i) \quad (16)$$

$$\text{Subject to : } \sum_{i=1}^5 m_i = B_m \quad (17)$$

$$c_2 + c_3 = B_{c1}, \quad c_4 + c_5 = B_{c2}, \quad (18)$$

where the decision variables are m_2, m_3, m_4, m_5, c_2 , and c_4 . Note that m_1, c_3 , and c_5 can be obtained from the constraints in (17) and (18), respectively. To obtain the optimal solution, we differentiate (15) with respect to each of the decision variables as follows:

$$\frac{\partial U_{tot}}{\partial m_2} = 0 \Rightarrow \frac{N_1}{B_m - m_2 - m_3 - m_4 - m_5} = \frac{N_2}{m_2 + c_2}, \quad (19)$$

$$\frac{\partial U_{tot}}{\partial m_3} = 0 \Rightarrow \frac{N_1}{B_m - m_2 - m_3 - m_4 - m_5} = \frac{N_3}{m_3 + B_{c1} - c_2 + B_{l1}}, \quad (20)$$

$$\frac{\partial U_{tot}}{\partial m_4} = 0 \Rightarrow \frac{N_1}{B_m - m_2 - m_3 - m_4 - m_5} = \frac{N_4}{m_4 + c_4}, \quad (21)$$

$$\frac{\partial U_{tot}}{\partial m_5} = 0 \Rightarrow \frac{N_1}{B_m - m_2 - m_3 - m_4 - m_5} = \frac{N_5}{m_5 + B_{c2} - c_4 + B_{l2}}, \quad (22)$$

$$\frac{\partial U_{tot}}{\partial c_2} = 0 \Rightarrow \frac{N_2}{m_2 + c_2} = \frac{N_3}{m_3 + B_{c1} - c_2 + B_{l1}}, \quad (23)$$

$$\frac{\partial U_{tot}}{\partial c_4} = 0 \Rightarrow \frac{N_4}{m_4 + c_4} = \frac{N_5}{m_5 + B_{c2} - c_4 + B_{l2}}. \quad (24)$$

Then, the optimal values of m_2, m_3, m_4, m_5, c_2 , and c_4 (that is, $m_2^*, m_3^*, m_4^*, m_5^*, c_2^*$, and c_4^*) are obtained by solving the above system of equations.

Obviously, (9), (10), (11), (12), (13), and (14) are the same as (19), (20), (21), (22), (23), and (24). This shows that the solution of the optimization formulation, which maximizes total network utility, is basically the Nash equilibrium obtained from the noncooperative game formulation described before.

4.2 Capacity Reservation

4.2.1 Prioritization among Different Types of Connections and Connection-Level Performance Measures

In a heterogeneous wireless access network, connections can be handed over between service areas with the same wireless access technologies (for example, from area 2 to area 4 in Fig. 1) or between areas with different access technologies (for example, from area 2 to area 3). The former is referred to as a horizontal handoff, the latter is referred to as a vertical handoff. To achieve prioritization among new and handoff connections, a portion of system capacity in a service area needs to be reserved for high-priority connections (for example, vertical and horizontal handoff connections). For this, a simple but effective guard channel [27] scheme was proposed in the literature. We adopt this guard channel concept to reserve a portion of system capacity for handoff connections. The parameters for this reservation scheme (that is, the reservation thresholds) are obtained from a bargaining game formulation among the different

types of connections. We consider a bargaining game for capacity reservation since all types of connections (that is, new and handoff connections) in a particular service area need to share the limited bandwidth (that is, capacity) offered by each network. The payoff for each type of connection is a function of connection blocking or dropping probability. In this case, the available capacity needs to be allocated to satisfy all types of connections in a service area. Note that if there are two types of connections (for example, new and horizontal handoff connections), the same bargaining game formulation is still applicable in which the number of players is two.

If A_a denotes the total bandwidth offered by all networks to service area a (for example, $A_2 = m_2 + c_2$), which is obtained from the network-level allocation, the total capacity (that is, number of connections) of service area a can be obtained from $C_a = \lfloor \frac{A_a}{R} \rfloor$, where R is the bandwidth requirement for a connection. In order to prioritize handoff connections over new connections, capacity $c_a^{(v)}$ and $c_a^{(h)}$ are reserved for vertical and horizontal handoff connections, respectively. In particular, a new connection can be accepted if the current number of ongoing connections is less than $\min(C_a - c_a^{(v)}, C_a - c_a^{(h)})$. A vertical and a horizontal handoff connection can be accepted if the current number of ongoing connection is less than $C_a - c^{(v)}$ and $C_a - c^{(h)}$, respectively. Based on an M/M/m/m queueing model, we can analytically obtain the average number of ongoing connections $\bar{n}_a$, the blocking probability for new connections $P_a^{(n)}$, and the dropping probabilities for vertical and horizontal handoff connections $P_a^{(v)}$ and $P_a^{(h)}$, respectively.

To quantify users' satisfaction as a function of the above connection-level performance measures, we use the following utility function [28]:

$$U = \frac{1}{1 - \sigma} \max(0, 1 - \sigma \exp(P_a)), \quad (25)$$

where σ ($0 < \sigma < 1$) is the parameter of the utility function. In particular, the larger the value of σ , the more sensitive is the utility to the connection blocking or connection dropping probability P_a .

4.2.2 Bargaining Game Formulation

To obtain the values of $c_a^{(v)}$ and $c_a^{(h)}$, which satisfy all the new connections, the vertical and horizontal handoff connections, we formulate a bargaining game in which different types of connections negotiate with each other to reach the equilibrium. A bargaining game can ensure fair payoffs for all the players while achieving prioritization among different types of connections. Note that a bargaining game is required for each service area.

The bargaining game formulation for capacity reservation can be described as follows:

- **Players.** The total number of players is three—an incoming new connection, a vertical handoff connection, and a horizontal handoff connection.
- **Strategies.** The strategies for the vertical handoff connection, the horizontal handoff connection, and

the new connection are the reserved capacities, that is, $c_a^{(v)}$, $c_a^{(h)}$, and $C_a - \max(c_a^{(v)}, c_a^{(h)})$, respectively.

- **Payoffs.** The payoff for each connection (that is, player) is the utility U obtained as a function of the corresponding connection-level performance. We use $U^{(n)}$, $U^{(v)}$, and $U^{(h)}$ to denote the payoffs for the new connection, the vertical and the horizontal handoff connections, respectively.

The payoff (that is, utility) for the players is given by $\Omega = \{(U^{(n)}, U^{(v)}, U^{(h)}) : 0 \leq U^{(n)}, U^{(v)}, U^{(h)} \leq 1\}$ (that is, feasible set). If an agreement among the players cannot be reached, the utility that the players will receive is given by the threat point $(\tilde{U}^{(n)}, \tilde{U}^{(v)}, \tilde{U}^{(h)})$. In particular, $(\tilde{U}^{(n)}, \tilde{U}^{(v)}, \tilde{U}^{(h)}) = (0, 0, 0)$ is the threat point for this game. This threat point represents that, if the game fails (and, hence, the resource reservations cannot be made), the players will start negotiation again.

The bargaining game is formulated as $\mathcal{F}(\Omega, \tilde{U}^{(n)}, \tilde{U}^{(v)}, \tilde{U}^{(h)}) = (U^{(n)*}, U^{(v)*}, U^{(h)*})$, where $(U^{(n)*}, U^{(v)*}, U^{(h)*})$ denotes the solution (equilibrium) [23] of the game $\mathcal{F}(\cdot)$. The Pareto optimality of the game defines an agreement such that one player cannot increase his utility without decreasing the utility of anyone of the other players. Specifically, the utility triplet $(U^{(n)}, U^{(v)}, U^{(h)})$ is Pareto optimal if there exists a utility triplet $(\hat{U}^{(n)}, \hat{U}^{(v)}, \hat{U}^{(h)})$ such that, if $U^{(n)} \geq \hat{U}^{(n)}$, $U^{(v)} \geq \hat{U}^{(v)}$, and $U^{(h)} \geq \hat{U}^{(h)}$, then $(\hat{U}^{(n)}, \hat{U}^{(v)}, \hat{U}^{(h)}) = (U^{(n)}, U^{(v)}, U^{(h)})$. The Pareto optimality can provide the candidate strategies (that is, $c_a^{(h)}$ and $c_a^{(v)}$) for which one of the players can achieve the highest utility. Then, we need the equilibrium of the game such that all of the players are satisfied with the utilities they receive. That is, the equilibrium of the bargaining game is the utility triplet $(U^{(n)*}, U^{(v)*}, U^{(h)*})$ such that [23]

$$(U^{(n)*}, U^{(v)*}, U^{(h)*}) = \arg \max_{U^{(n)}, U^{(v)}, U^{(h)}} \left(U^{(n)} - \tilde{U}^{(n)} \right) \left(U^{(v)} - \tilde{U}^{(v)} \right) \left(U^{(h)} - \tilde{U}^{(h)} \right). \quad (26)$$

Since the size of the strategy space (that is, $(c_a^{(h)}, c_a^{(v)})$) is small, we obtain the solution of this bargaining game by enumeration.

5 CONNECTION-LEVEL BANDWIDTH ALLOCATION AND ADMISSION CONTROL

While network-level bandwidth allocation and capacity reservation are performed on a long-term basis based on the average number of ongoing connections in a service area at the steady state, connection-level allocation is performed in each service area upon arrival/departure of a connection (that is, on a short-term basis). When a connection arrives, the amount of bandwidth to be offered to an incoming connection is determined and also the admission control procedure is invoked. When a connection departs a service area, the bandwidth released from the connection is distributed among the ongoing connections so that the resource utilization can be maximized.

5.1 Noncooperative Game for Connection-Level Bandwidth Allocation

For connection-level bandwidth allocation, we formulate a noncooperative game among the networks available in a particular service area (for example, the WMAN and cellular network in area 2 and the WMAN, cellular network, and WLAN in area 4 in Fig. 1). These networks compete with each other to offer bandwidth to an incoming connection (that is, new connection, vertical handoff connection, or horizontal handoff connection) in order to maximize their profits (that is, utility), and also, all the networks are satisfied with the solution of the game. The decision on the amount of offered bandwidth to a new connection by a network depends on the actions taken by other networks. In particular, one network will receive small utility from a new connection if other networks offer a large amount of bandwidth to that connection. However, if a particular network has many ongoing connections, offering a large amount of bandwidth to a new connection will degrade the network utility. We consider this performance degradation as a cost of offering bandwidth to a new connection. This conflicting situation is modeled as a noncooperative game.

5.1.1 Formulation of the Game

The number of networks providing service to a new connection in a particular service area is M ($M \in \{2, 3\}$). The supplied service here is the bandwidth and a new connection is a customer in the market. Based on this model, the strategic form of the game can be defined as follows:

- **Players.** The players in this game are the networks available in a particular service area.
- **Strategies.** The strategy of each of the players is the amount of offered bandwidth to an incoming connection (denoted by p_i for network i) which is non-negative.
- **Payoffs.** The payoff for each player is the profit (that is, revenue minus cost) of network i (denoted by ϕ_i) in offering bandwidth to an incoming connection.

The revenue (that is, utility) of network i is computed from the surplus utility over that due to the allocations by other networks and can be obtained as follows:

$$V_i^{(con)}(\mathbb{P}) = w \left[\log \left(\alpha \sum_j p_j \right) - \log \left(\alpha \sum_{j \neq i} p_j \right) \right], \quad (27)$$

where $\mathbb{P}$ is the set containing the amount of bandwidth offered by all networks to an incoming connection (that is, pure strategies of all networks). This set is defined as follows: $\mathbb{P} = \{\dots p_j \dots p_i \dots\}$.

Since the bandwidth offered to an incoming connection must be taken from the ongoing connections, for network i , the cost of offering bandwidth p_i can be considered as a loss in total utility. This cost can be calculated as follows:

$$F_i(p_i) = \sum_x \left(U_{i,x}^{(con)}(\mathbb{Q}^{(x)}(0)) - U_{i,x}^{(con)}(\mathbb{Q}^{(x)}(p_i)) \right), \quad (28)$$

where $U_{i,x}^{(con)}(\cdot)$ is the utility gained by network i from connection x (as in (1)) and $\mathbb{Q}^{(x)}(p_i)$ represents a set the elements of which denote the amount of bandwidth allocated to connection x . Specifically, p_i is the amount of bandwidth offered to an incoming connection by network i . Note that the bandwidth allocated to the ongoing connections in a service area depends on the network-level bandwidth allocation. The set $\mathbb{Q}^{(x)}(p_i)$ can be defined as follows:

$$\mathbb{Q}^{(x)}(p_i) = \left\{ \dots \frac{D_k q_k^{(x)}}{\sum_x q_k^{(x)}} \dots \frac{(D_i - p_i) q_i^{(x)}}{\sum_x q_i^{(x)}} \dots \right\}, \quad (29)$$

where $q_i^{(x)}$ is the bandwidth currently allocated to connection x in network i and D_i is the bandwidth allocated by network i in a particular service area (that is, $D_i \in \{m_j, c_k, B_l\}$, B_l is the WLAN bandwidth in a service area). In particular, the first term of the cost function in (28) is the total utility of network i gained from the ongoing connections and the second term is the total utility of network i after offering bandwidth p_i to an incoming connection. The profit of network i in offering bandwidth p_i to an arriving connection is then defined as follows:

$$\phi_i(\mathbb{P}) = V_i^{(con)}(\mathbb{P}) - f_i F_i(p_i), \quad (30)$$

where f_i (which is a positive real number) denotes the weight of cost function for network i .

5.1.2 Nash Equilibrium of the Noncooperative Game

The Nash equilibrium is obtained by using the best response function, which is the best strategy of one player given other players' strategies. The best response function for network i (in the connection-level) given the amount of bandwidth offered by other networks p'_j (for $j \neq i$) is defined as follows:

$$BR_i^{(con)}(\mathbb{P}') = \arg \max_{p_i} \phi_i(\mathbb{P}'), \quad (31)$$

where set $\mathbb{P}'$ contains the amount of bandwidth offered by network j ($j \neq i$) and it can be defined as $\mathbb{P}' = \{\dots p'_j \dots\}$. In this case, set $\mathbb{P}$ can be obtained as follows: $\mathbb{P} = \mathbb{P}' \cup \{p_i\}$.

For a service area with three wireless access networks, we can express the best response function, given other networks' strategies p'_j and p'_k , as follows:

$$BR_i^{(con)}(\{p'_j, p'_k\}) = \arg \max_{p_i} \phi_i(\{p'_j, p'_k, p_i\}). \quad (32)$$

The set $\mathbb{P}^* = \{p_j^*, p_k^*, p_i^*\}$ denotes the Nash equilibrium of this game if and only if

$$p_i^* = BR_i^{(con)}(\{p_j^*, p_k^*\}), \quad p_j^* = BR_j^{(con)}(\{p_i^*, p_k^*\}), \quad p_k^* = BR_k^{(con)}(\{p_i^*, p_j^*\}). \quad (33)$$

Similarly, in the presence of two networks, the best response function can be expressed as follows: $BR_i^{(con)}(\{p'_j\}) = \arg \max_{p_i} \phi_i(\{p'_j, p_i\})$. The set $\mathbb{P}^* = \{p_j^*, p_i^*\}$ denotes Nash equilibrium of this bandwidth allocation game in the two-network case if $p_i^* = BR_i^{(con)}(\{p_j^*\})$, $p_j^* = BR_j^{(con)}(\{p_i^*\})$.

Observation 1. The Nash equilibrium of the above noncooperative game is unique.

Proof. To prove the uniqueness of the Nash equilibrium, we particularly consider the case with two networks—WMAN and cellular network. First, we obtain the best response function by differentiating the profit function in (30) with respect to p_i given p'_j , where $\mathbb{P} = \{p_i, p'_j\}$. The profit function can be expressed in a general form as follows:

$$\begin{aligned} \phi_i(\{p_i, p'_j\}) = & w \left[\log(\alpha(p_i + p'_j)) - \log(\alpha p'_j) \right. \\ & - f_i \left(\sum_y \log \left(\alpha \frac{D_i q_i^{(y)}}{\sum_x q_i^{(x)}} \right) \right. \\ & \left. \left. - \sum_y \log \left(\alpha \frac{(D_i - p'_j) q_i^{(y)}}{\sum_x q_i^{(x)}} \right) \right) \right]. \end{aligned} \quad (34)$$

Now,

$$\frac{\partial \phi_i(\{p_i, p'_j\})}{\partial p_i} = 0 \quad (35)$$

gives

$$p_i^* = \frac{D_i - f_i n p'_j}{1 + f_i n}, \quad (36)$$

where n is the current number of ongoing connections in a particular service area. Similarly, for the best response function of network j , we have

$$p_j^* = \frac{D_j - f_j n p'_i}{1 + f_j n}. \quad (37)$$

Therefore, the best response function for each of these two networks is a linear function of the strategy adopted by the other network.

The Nash equilibrium is unique if and only if the slopes of these linear best response functions are unequal. We show this by reformatting (37) as follows:

$$p'_i = \frac{D_j - p_j^*(1 + f_j n)}{f_j n}. \quad (38)$$

The slopes of p_i^* and p'_i , given p'_j and p_j^* , are $-\frac{f_i n}{1 + f_i n}$ and $-\frac{1 + f_j n}{f_j n}$, respectively. Since

$$\frac{f_i n}{1 + f_i n} = \frac{1 + f_j n}{f_j n} \quad (39)$$

gives $f_j n + f_i n = -1$, which is impossible (since f_j, f_i , and n must be positive to make the cost in (30) a negative number), by contradiction, the uniqueness of the Nash equilibrium is established. A similar procedure can be used to show that the Nash equilibrium is unique in a three-network case. This completes the proof. $\square$

5.1.3 A Heuristic Search Algorithm to Compute the Nash Equilibrium

Since obtaining the Nash equilibrium may be computationally intensive (depending on the number of connections

involved and the type of the revenue and the cost functions), we apply a heuristic search algorithm that is easy to implement and also applicable for a wide range of utility functions. In this case, we minimize an objective function that depends on the strategies adopted by other networks.

Mathematically, the Nash equilibrium is given as follows: $\mathbb{P}^* = \{p_j^*, p_k^*, p_i^*\}$, where $\{p_j^*, p_k^*, p_i^*\} = \arg \min_{p_j, p_k, p_i} \Psi(p_j, p_k, p_i)$, where

$$\begin{aligned} \Psi(p_j, p_k, p_i) = & \left| p_i - BR_i^{(con)}(\{p_j, p_k\}) \right| \\ & + \left| p_j - BR_j^{(con)}(\{p_i, p_k\}) \right| \\ & + \left| p_k - BR_k^{(con)}(\{p_i, p_j\}) \right| \end{aligned}$$

in which the objective function $\Psi(\cdot)$ is defined as the difference between the best responses and the strategies adopted by other networks.

However, in a three-network case, given the strategies of two networks, we can obtain the best response for the other network; therefore, the number of decision variables is reduced to two. Given strategies p_j and p_k , the objective function can be computed as in Algorithm 1. A standard search algorithm can be used to obtain the solution. We use the *Nelder-Mead* direct search method [29] in this paper.

Algorithm 1 Objective function for the search algorithm.

Input: p_j and p_k

- 1: $p'_i \leftarrow BR_i^{(con)}(\{p_j, p_k\})$
- 2: $p'_j \leftarrow BR_j^{(con)}(\{p'_i, p_k\})$
- 3: $p'_k \leftarrow BR_k^{(con)}(\{p'_i, p'_j\})$
 {compute objective function}
- 4: $\Psi \leftarrow |p'_j - p_j| + |p'_k - p_k|$

Output: Ψ

5.1.4 Iterative Algorithm to Compute the Nash Equilibrium

In Algorithm 1, we observe that the objective function can be computed from auxiliary variables p'_j and p'_k . Therefore, we introduce an iterative algorithm in which we use these auxiliary variables again to obtain new variables in the next iteration (Algorithm 2). With starting points p_j^{st} and p_k^{st} , the algorithm iterates until the difference between variables in the previous iteration (that is, p_j^{pr} and p_k^{pr}) and in the current iteration (that is, p_j^{cu} and p_k^{cu}) becomes less than the threshold th (for example, $th = 10^{-4}$).

Algorithm 2 Iterative algorithm.

Input: p_j^{st}, p_k^{st}

- 1: $p_j^{pr} \leftarrow p_j^{st}, p_k^{pr} \leftarrow p_k^{st}$ {initialize variables in the current iteration}
- 2: **repeat**
- 3: $p_i^{cu} \leftarrow BR_i^{(con)}(\{p_j^{pr}, p_k^{pr}\})$
- 4: $p_j^{cu} \leftarrow BR_j^{(con)}(\{p_i^{cu}, p_k^{pr}\})$
- 5: $p_k^{cu} \leftarrow BR_k^{(con)}(\{p_i^{cu}, p_j^{cu}\})$
- 6: $\epsilon \leftarrow |p_j^{cu} - p_j^{pr}| + |p_k^{cu} - p_k^{pr}|$ {compute the difference}
- 7: $p_j^{pr} \leftarrow p_j^{cu}, p_k^{pr} \leftarrow p_k^{cu}$ {update variables}
- 8: **until** $\epsilon \leq th$ {termination criterion}

Output: $p_i^{cu}, p_j^{cu}, p_k^{cu}$

Observation 2. Given a starting point, the iterative algorithm converges.

Proof. Since the solutions of (36) and (37) can be either both nonzero positive (**Case I**) or one negative (or zero) and the other positive (**Case II**),² we provide proofs for these two separate cases.

Case I. The iterative algorithm $p_i(t) = \mathcal{F}(p_i(t-1))$, where $p_i(t)$ is the strategy at iteration t , converges if the following conditions are satisfied [30]: First, a solution point (that is, Nash equilibrium) must exist. Second, the function $\mathcal{F}(p)$ should have the following three properties: positivity (that is, $\mathcal{F}(p) > 0$), monotonicity (that is, $p > p' \Rightarrow \mathcal{F}(p) > \mathcal{F}(p')$), and scalability (that is, $\beta > 1 \Rightarrow \mathcal{F}(\beta p) < \beta \mathcal{F}(p)$).

For the proposed iterative algorithm, we have proved that the Nash equilibrium is unique (observation 1). For monotonicity, we rewrite (36) as follows:

$$\begin{aligned} p_i(t) &= \mathcal{F}(p_i(t-1)) \\ &= \frac{1}{1 + f_j n} \left(D_i - f_i n \left(\frac{D_j - f_j n p_i(t-1)}{1 + f_j n} \right) \right). \end{aligned} \quad (40)$$

It is easy to show that, if $p > p'$, then

$$\frac{-D_j + f_j n p}{1 + f_j n} > \frac{-D_j + f_j n p'}{1 + f_j n} \quad (41)$$

$$\begin{aligned} &\Rightarrow \frac{1}{1 + f_i n} \left(D_i - f_i n \left(\frac{D_j - f_j n p}{1 + f_j n} \right) \right) \\ &> \frac{1}{1 + f_i n} \left(D_i - f_i n \left(\frac{D_j - f_j n p'}{1 + f_j n} \right) \right) \end{aligned} \quad (42)$$

$$\Rightarrow \mathcal{F}(p) > \mathcal{F}(p'). \quad (43)$$

For scalability, since $\beta > 1$, we have

$$D_i - \frac{f_i n D_j}{1 + f_j n} < \beta \left(D_i - \frac{f_i n D_j}{1 + f_j n} \right) \quad (44)$$

$$\begin{aligned} &\Rightarrow \frac{1}{1 + f_i n} \left(D_i - f_i n \left(\frac{D_j - f_j n \beta p}{1 + f_j n} \right) \right) \\ &< \beta \frac{1}{1 + f_i n} \left(D_i - f_i n \left(\frac{D_j - f_j n p}{1 + f_j n} \right) \right) \end{aligned} \quad (45)$$

$$\Rightarrow \mathcal{F}(\beta p) < \beta \mathcal{F}(p). \quad (46)$$

These properties also hold for $p_j(t) = \mathcal{G}(p_j(t-1))$.

Case II. We show that, if the amount of allocated bandwidth to one player becomes negative, the bandwidth allocated to the other player converges to a fixed positive solution. Since the amount of bandwidth cannot be negative in a real scenario, in the algorithm, we set it to zero.

Without loss of generality, assuming that $p_i(t-1) > 0$, $p_i(t) < 0$, and $p_j(t) > 0$, we have

$$p_i(t) = 0, \quad \text{when} \quad \frac{D_i - f_i n p_j(t)}{1 + f_i n} < 0. \quad (47)$$

Then,

$$p_j(t+1) = \mathcal{G}(p_i(t) = 0) = \frac{D_j - f_j n \times 0}{1 + f_j n}. \quad (48)$$

2. Both the solutions of (36) and (37) cannot be negative or zero.

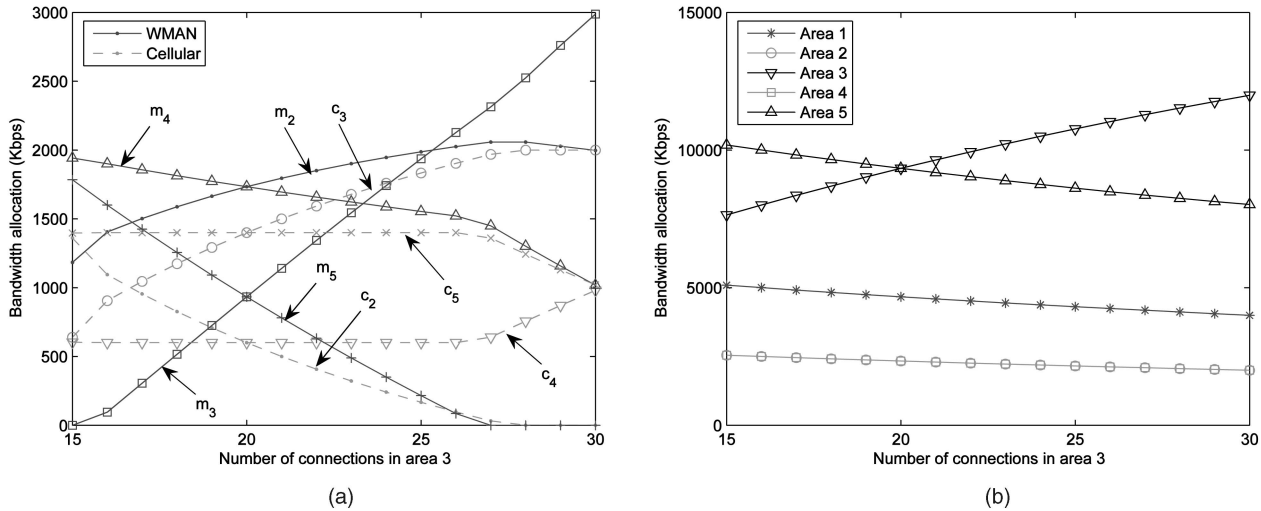


Fig. 3. (a) Bandwidth allocated by different networks to each service area and (b) the total amount of bandwidth allocated to each service area.

Therefore, $p_j(t+1) > p_j(t)$ and, in general, $p_i(t+z) < 0$, $\forall z > 0$. Hence, the iterative algorithm $\mathcal{G}(p_i(t))$ will converge to

$$p_j^* = p_j(t+1) = \mathcal{G}(p_i(t) = 0). \quad (49)$$

A similar procedure can be used to show that the algorithm converges in a three-network case. This completes the proof. $\square$

5.2 Bandwidth Distribution

When a connection departs a service area, the bandwidth released from the departing connection is distributed among the ongoing connections. The distribution of bandwidth is based on the current amount of allocated bandwidth to the ongoing connections as follows:

$$q_x = q_x + \frac{(\max(q_y) - q_x)\hat{q}}{\sum_x (\max(q_y) - q_x)}, \quad (50)$$

where q_x is the bandwidth allocated to the ongoing connection x and $\hat{q}$ is the bandwidth released by the departing connection. Note that, if all of the connections have the same bandwidth, $\hat{q}$ is distributed equally among the ongoing connections.

5.3 Admission Control Algorithm

When a user initiates a new connection, the information on the required bandwidth is sent to the central controller, which computes the offered bandwidth by each network. In this case, either the search algorithm or the iterative algorithm is used to obtain p_i^* , p_j^* , and p_k^* , which denote the amount of bandwidth offered by network i , j , and k in a particular service area, respectively, to a new connection. Since the admission control algorithm ensures that the requested bandwidth of an incoming connection is satisfied when it is admitted, the following condition is checked:

$$p_i^* + p_j^* + p_k^* \geq R, \quad (51)$$

where R is the bandwidth requirement of a new connection.

If the incoming connection is a newly initiated connection, the admission control procedure checks whether the

total number of ongoing connections in a particular area is less than $C_a - \min(c_a^{(v)}, c_a^{(h)})$. If this is true, an arriving connection is accepted. Similarly, the admission control procedure checks with threshold $c_a^{(v)}$ and $c_a^{(h)}$ for vertical and horizontal handoff connections, respectively.

6 PERFORMANCE EVALUATION

6.1 Parameter Setting

We consider the service areas shown in Fig. 1. In case of IEEE 802.11 WLAN, the channel rate is 11 Mbps and the maximum saturation throughput achieved through EBA in a WLAN is 7 Mbps [20]. For the CDMA cellular wireless access, the transmission bandwidth is assumed to be 5 MHz. We assume that the ratio of bit energy and noise-plus-interference power spectral density at the receiver is 8.17 dB, which corresponds to a bit-error rate of 10^{-4} . The total transmission rate in each CDMA cell is 2 Mbps. For the IEEE 802.16-based wireless access, the transmission rate is 10 Mbps in a single cell. Note that the subscripts wm , ce , and wl are used to denote the parameters corresponding to WMAN, cellular network, and WLAN, respectively.

The parameters for the network utility function are set as follows: $w = 1$ and $\alpha = 0.7$. The parameter for the utility function for a connection (σ) is set to 0.7, 0.8, and 0.9, respectively, for a new connection, a vertical handoff connection, and a horizontal handoff connection. With these values of σ , the utilities for a new connection, a vertical handoff connection, and a horizontal handoff connection become zero if the corresponding blocking and dropping probabilities become higher than 0.3, 0.1, and 0.05, respectively.

6.2 Network-Level Allocation

Fig. 3a shows the solution obtained from the network-level allocation, that is, the amount of bandwidth allocated by each network to each of the service areas. In this case, the average number of connections in area 1, 2, 4, and 5 is 10, 5, 5, and 20, respectively, whereas that of area 3 is varied. As the number of connections in area 3 increases, the amount of bandwidth corresponding to this area (that is, m_3 and c_3

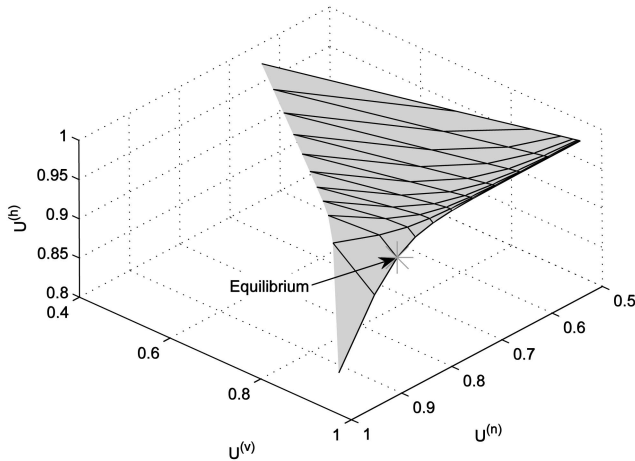


Fig. 4. Pareto optimality and equilibrium of the bargaining game for capacity reservation.

offered by WMAN and cellular network) increases accordingly. Since the total bandwidth of the cellular network is limited, the bandwidth allocated by the cellular network to area 2 (that is, c_2) decreases significantly. However, to be fair to the connections in area 2, WMAN tries to allocate a larger amount of bandwidth to this area (that is, m_2 increases). On the other hand, WMAN needs to reduce the amount of bandwidth allocated to other areas (that is, m_1 , m_4 , and m_5). In this case, since area 5 is serviced by WLAN for which the available bandwidth is large (for example, 7 Mbps), m_5 decreases at a rate higher than that for each of m_1 and m_4 .

Note that the bandwidth allocation by the cellular network in-service area 4 is mostly unaffected even though the number of connections in area 3 increases. However, when the number of connections in area 3 becomes very large (for example, more than 27 connections), since WMAN cannot contribute bandwidth to area 5 (that is, m_5), the cellular network needs to alter its allocations for area 4 and area 5 (that is, c_4 and c_5 , respectively).

Fig. 3b shows the amount of bandwidth allocated to each area. As expected, the total amount of bandwidth allocated to area 3 increases as the number of connections increases. In this case, since the bandwidth from WMAN allocated to area 5 is taken away and given to area 3, the total amount of bandwidth allocated to area 5 decreases. We observe that both area 2 and area 4 receive an equal amount of bandwidth since they serve the same number of connections. Similarly, when the number of connections in both area 3 and area 5 is 20 (in Fig. 3b), the same amount of bandwidth is allocated to each of these service areas. These results show that the noncooperative game provides fair bandwidth allocation at the network level.

6.3 Capacity Reservation

Fig. 4 shows the Pareto optimality and the equilibrium of the capacity reservation for area 3 in which the average arrival rate for new, vertical handoff, and horizontal handoff connections is 2.4, 1.2, and 0.6, respectively, and the average connection holding time is 10 minutes. It is expected that, while one player can increase his payoff, the payoff of another player must be decreased, since the threshold setting will affect the connection blocking/dropping probabilities of all the players. In this case, at the equilibrium, the capacity reserved for vertical handoff and horizontal handoff connections is 2 and 1, respectively.

6.4 Connection-Level Allocation

6.4.1 Best Response Functions

Fig. 5a shows the best responses for WMAN and cellular network under different strategies in-service area 2. The amount of bandwidth allocated by WMAN and cellular network to this area is 1,400 and 1,100 Kbps, respectively (which corresponds to the average number of connections of 16 in area 3 in Fig. 3a). The Nash equilibrium is located at the point where the best responses of WMAN and cellular network intersect. The equilibrium varies with the number of ongoing connections. The smaller number of ongoing

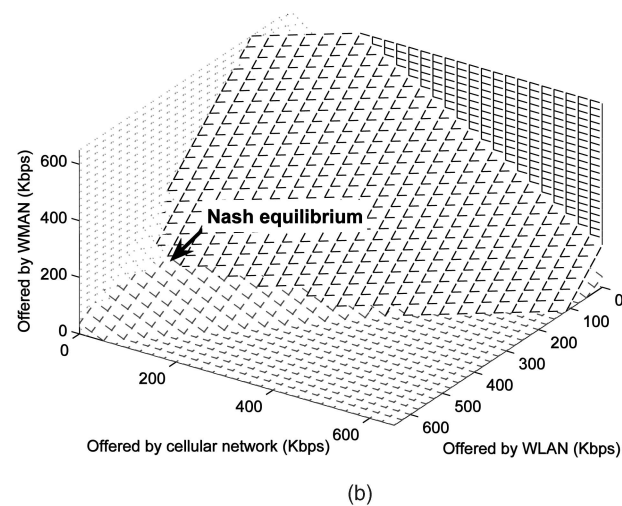
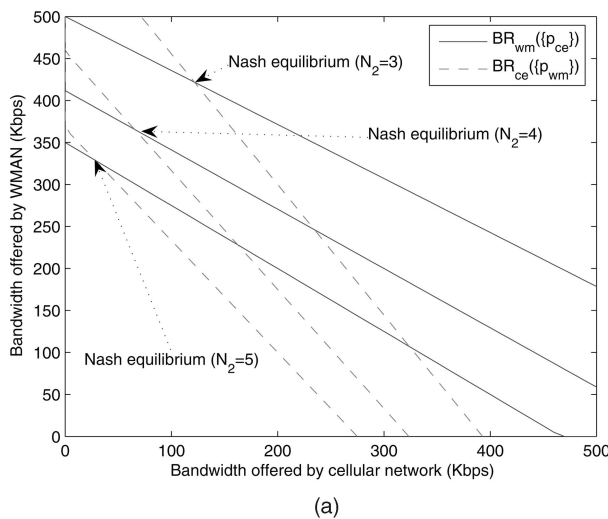


Fig. 5. Best response functions of (a) WMAN and cellular network in-service area 2, and (b) WMAN, cellular network, and WLAN in-service area 3. Authorized licensed use limited to: Indian Statistical Institute Kolkata. Downloaded on February 06, 2025 at 20:17:38 UTC from IEEE Xplore. Restrictions apply.

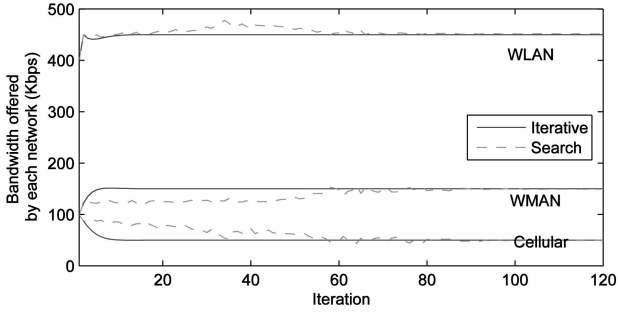


Fig. 6. Comparison of speed of convergence between the iterative and the search algorithms.

connections result in a larger amount of bandwidth offered to an arriving connection, and vice versa.

Fig. 5b shows the best response functions for WMAN, cellular network, and WLAN in service area 3. The amount of bandwidth allocated by WMAN, cellular network, and WLAN to this area is 2,100, 1,900, and 2,400 Kbps, respectively, and the number of ongoing connections is assumed to be 5 in this service area. In this case, the best response function of each player is a plane, and the Nash equilibrium is located at the intersection of these planes. Since the WLAN and the cellular network have the largest and the smallest amount of available bandwidth, the bandwidth offered to an incoming connection from WLAN and cellular network is the highest and the lowest, respectively.

6.4.2 Iterative and Heuristic Search Algorithms

Fig. 6 shows the speed of convergence of the iterative and the heuristic search algorithms to obtain the Nash equilibrium in a three-player noncooperative game. The starting point for both the algorithms is set to 100 Kbps for both WMAN and cellular network. Although both the algorithms achieve the same solution, we observe that the iterative algorithm can reach the final solution much faster (that is, within 15 iterations in Fig. 6) and more smoothly than the search algorithm. Therefore, the iterative algorithm is superior to the search algorithm in terms of both stability and efficiency.

6.4.3 Bandwidth Adaptation

Fig. 7a shows variations in the amount of bandwidth offered to a new connection in area 3 under a different number of ongoing connections. As expected, when the number of connections in this area is small, an incoming connection will receive a large amount of bandwidth. This bandwidth decreases as the number of ongoing connections increases. Also, bandwidth offered to an arriving connection by WLAN is the largest since WLAN has the highest available bandwidth. We observe that the amount of available bandwidth in WLAN has a significant impact on the amount of bandwidth offered to an arriving connection (Fig. 7b).

6.5 Performance of Admission Control

We assume that the bandwidth requirement for every connection in the network is 200 Kbps. That is, if the

connection-level bandwidth allocation cannot allocate bandwidth larger than 200 Kbps to an incoming connection, that connection is rejected. The arrival rates of new, vertical handoff, horizontal handoff connections for area 1, 2, 4, and 5 are (1.4, 0.7, 0.35), (0.4, 0.2, 0.1), (0.6, 0.3, 0.15), and (1.2, 0.6, 0.3), respectively. The connection arrival rates in area 3 are denoted by $\gamma(2, 1, 0.5)$, where γ is the traffic intensity.³

Fig. 8a shows variations in the average amount of bandwidth allocated to a connection in each service area. The bandwidth allocated to a connection in service area 3 becomes the highest in the network when there are a small number of ongoing connections (for example, when traffic intensity is 0.4-0.5). In this case, the WLAN can offer a large amount of bandwidth to a connection in service area 3. When the traffic intensity increases, connections in area 1 and area 3 receive a slightly lower amount of bandwidth than other areas since traffic load in both these areas is higher than the load in other areas. When the traffic load in area 3 increases, the average amount of bandwidth allocated to a connection in most of the service areas decreases due to the load balancing achieved through the game-theoretic bandwidth allocation. However, the connection-level allocation in area 5 is not affected by traffic load in area 3 since, in area 5, the WLAN contributes most of the bandwidth to the connections.

Figs. 8b, 9a, and 9b show variations in new connection blocking and connection dropping probabilities for vertical and horizontal handoff connections. As expected, traffic load in area 3 affects the connection-level performances in area 1, 2, and 4 (but not area 5). Due to the capacity reservation, in each service area, the handoff connection blocking probability is smaller than the new connection blocking probability. In this case, connections in area 2 and area 4 experience high blocking and dropping probabilities since the cellular networks in these areas have to share the bandwidth with the connections in area 3 and area 5. Also, the WMAN cannot contribute a large amount of bandwidth to area 2 and area 4 since it needs to serve the connections in area 1 in which only WMAN service is available.

6.6 Summary of the Observations

The performance evaluation results can be summarized as follows:

- The network-level bandwidth allocation tries to allocate bandwidth to the service areas from the different access networks in a fair manner.
- Combined with network-level allocation, capacity reservation can be used to obtain the reservation thresholds for vertical and horizontal handoff connections so that connection blocking and dropping probabilities are maintained below the target level.
- Connection-level bandwidth allocation (that is, bandwidth allocation in a short-term basis) is required to adapt with the traffic fluctuation in a service area, as well as the variation in the available bandwidth from the different access networks.
- The iterative algorithm for connection-level bandwidth allocation can converge to the solution quickly

3. The traffic load corresponding to new vertical handoff and horizontal handoff connections is obtained by multiplying γ with 2, 1, and 0.5, respectively.

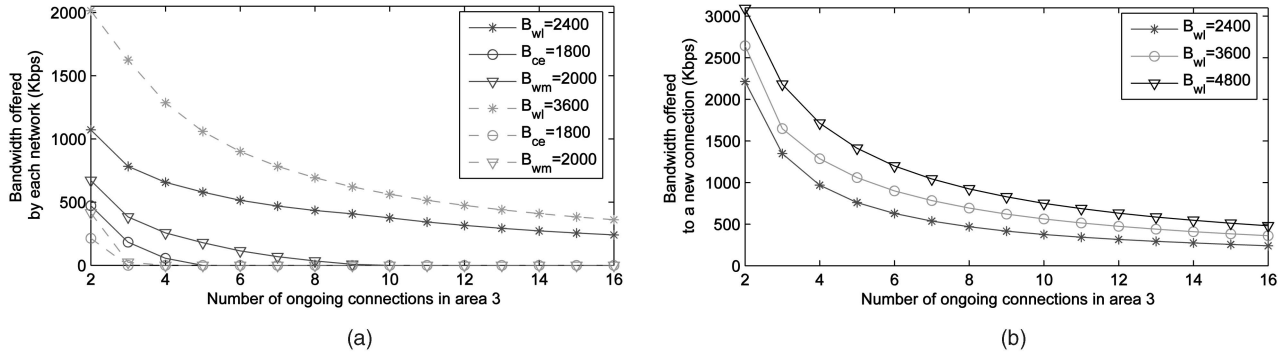


Fig. 7. (a) The amount of bandwidth offered by each network and (b) the total amount of bandwidth received by a new connection.

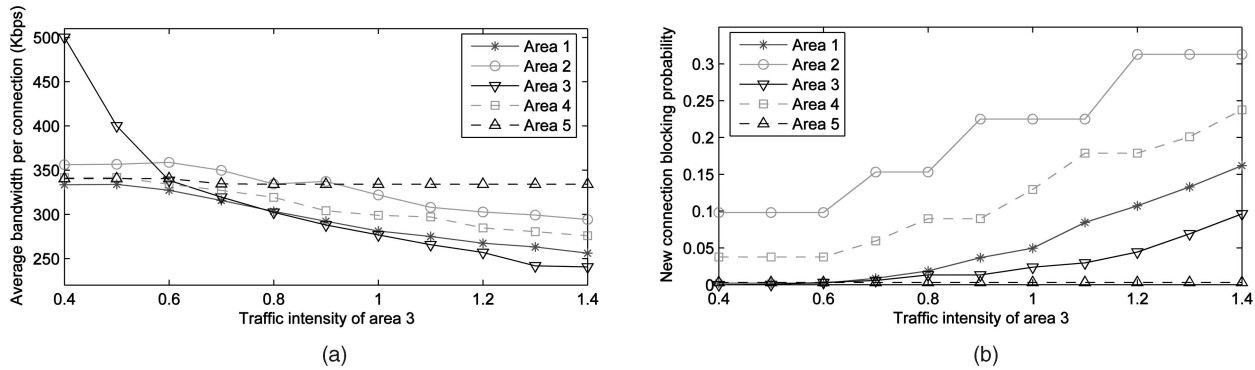


Fig. 8. (a) Average amount of allocated bandwidth per connection and (b) new connection blocking probability.

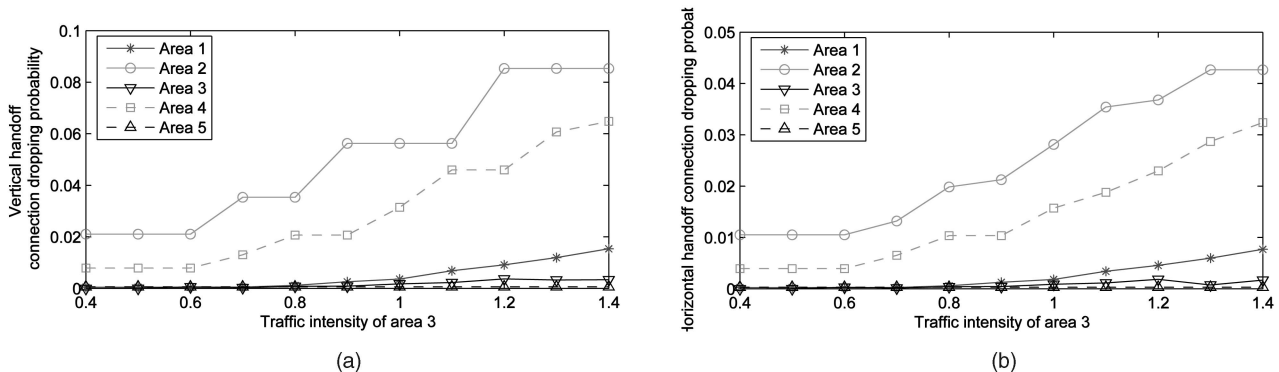


Fig. 9. (a) Horizontal and (b) vertical handoff connection dropping probability.

and, therefore, would be more suitable for online execution.

- In a heterogeneous wireless access environment, admission control is required not only to maintain the performances of the ongoing connections at the desired level but also to prioritize handoff connections over new connections.

7 CONCLUSION

We have presented a game-theoretic framework for RRM in heterogeneous wireless access networks consisting of WMAN, cellular networks, and WLANs. This framework provides a fair resource allocation in the different service areas while satisfying both the service providers' and the users' requirements. Also, it can adapt to both long-term

and short-term variations of network resources and traffic load conditions. The performances of the different components of this framework, namely, network-level bandwidth allocation, capacity reservation, connection-level bandwidth allocation, and admission control have been analyzed. The proposed framework would be useful for RRM and protocol engineering in the evolving heterogeneous wireless access networks.

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